

# FUNDAMENTALS OF JAVA



# CORE PRINCIPLE



WORA- “Write Once Run  
Anywhere”

Bytecode

But it works on my machine



# JAVA SYNTAX

| CODE                 | MEANING                                                                |
|----------------------|------------------------------------------------------------------------|
| <b>public</b>        | JVM can access this method from anywhere.                              |
| <b>static</b>        | Belongs to the class, not an object (so it runs without creating one). |
| <b>void</b>          | Method doesn't return any value.                                       |
| <b>main</b>          | Program's starting point.                                              |
| <b>String[] args</b> | Stores command-line inputs                                             |

# HelloWorld.java

J HelloWorld.java x

J HelloWorld.java > HelloWorld

```
1 public class HelloWorld{  
    Run | Debug  
2     public static void main(String[] args) {  
3         System.out.println("Hello World!");  
4     }  
5 }
```

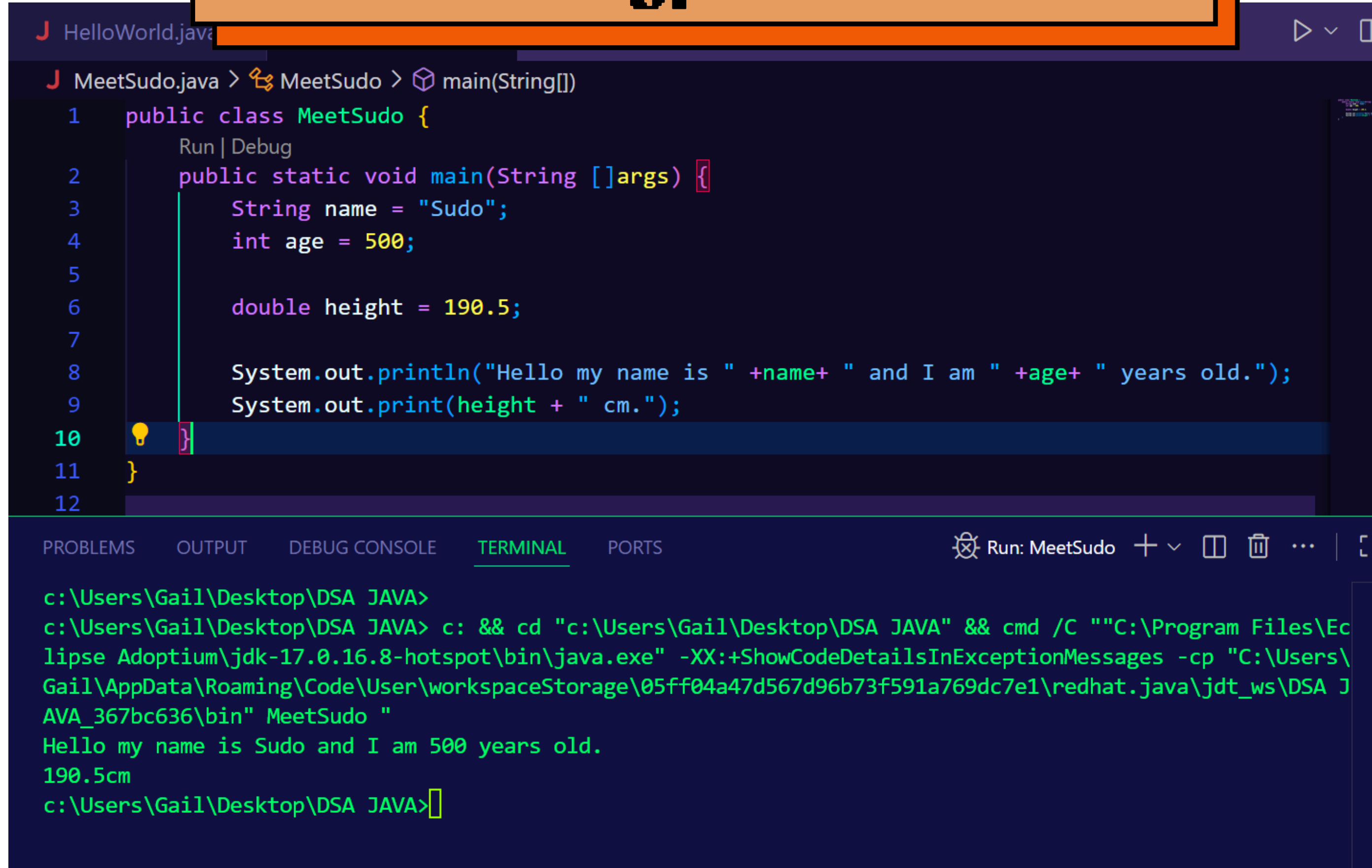
PROBLEMS OUTPUT TERMINAL ...

Run: HelloWorld + v [icon] [icon] ... [icon] [icon]

```
e\User\workspaceStorage\05ff04a47d567d96b73f591a769dc7e1\redhat.java\jdt_ws\DS  
A JAVA_367bc636\bin" HelloWorld "  
Hello World!
```

```
c:\Users\Gail\Desktop\DSA JAVA>
```

# String, double, int data types



The screenshot shows an IDE window with a Java file named `MeetSudo.java`. The code defines a `MeetSudo` class with a `main` method. The `main` method declares three variables: `String name = "Sudo";`, `int age = 500;`, and `double height = 190.5;`. It then prints a message using `System.out.println` and `System.out.print`. The IDE interface includes a toolbar with 'Run' and 'Debug' buttons, and a bottom panel with tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'TERMINAL' tab is active, showing the command to run the program and its output.

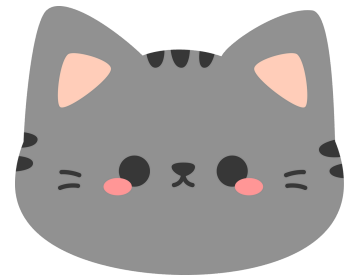
```
1 public class MeetSudo {  
2     public static void main(String []args) {  
3         String name = "Sudo";  
4         int age = 500;  
5  
6         double height = 190.5;  
7  
8         System.out.println("Hello my name is " + name + " and I am " + age + " years old.");  
9         System.out.print(height + " cm.");  
10    }  
11 }  
12
```

Run: MeetSudo

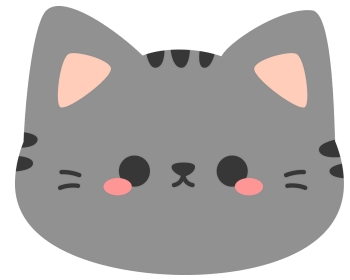
```
c:\Users\Gail\Desktop\DSA JAVA>  
c:\Users\Gail\Desktop\DSA JAVA> c: && cd "c:\Users\Gail\Desktop\DSA JAVA" && cmd /C ""C:\Program Files\Eclipse Adoptium\jdk-17.0.16.8-hotspot\bin\java.exe" -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\Gail\AppData\Roaming\Code\User\workspaceStorage\05ff04a47d567d96b73f591a769dc7e1\redhat.java\jdt_ws\DSA JAVA_367bc636\bin" MeetSudo "  
Hello my name is Sudo and I am 500 years old.  
190.5cm  
c:\Users\Gail\Desktop\DSA JAVA>
```



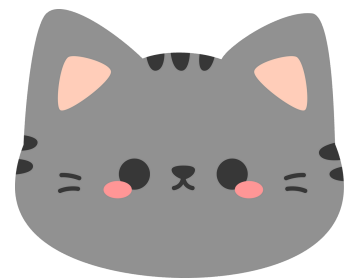
# Arrays



**Fixed list — Basic Array**



**Empty first, fill later — Declared Array**



**Growing list — ArrayList**

# Fixed ListArrays

## Array to string conversion

MeetSudo.java > MeetSudo

```
1  import java.util.Arrays;
2
3  public class MeetSudo {
4      public static void main(String []args) {
5          String name = "Sudo";
6          int age = 500;
7
8          double height = 190.5;
9
10         String [] foods = {"Pizza", "Burger", "Pasta", "Salad"};
11
12         System.out.println("Hello my name is " +name+ " and I am " +age+ " years old.");
13         System.out.println(height + " cm.");
14         System.out.println("My favorite foods are:" +Arrays.toString(foods));
15
16     }
17 }
18
19
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: MeetSudo + - [ ] ... [ ]

```
8-hotspot\bin\java.exe" -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\Gail\AppData\Roaming\Code\User\workspaceStorage\05ff04a47d567d96b73f591a769dc7e1\redhat.java\jdt_ws\DSA_JAVA_367bc636\bin" MeetSudo "
Hello my name is Sudo and I am 500 years old.
190.5 cm.
My favorite foods are:[Pizza, Burger, Pasta, Salad]
```

# Fixed ListArrays

Without import java.util.Arrays

```
J MeetSudo.java > MeetSudo
1 public class MeetSudo {
    Run | Debug
2     public static void main(String []args) {
3         String name = "Sudo";
4         int age = 500;
5
6         double height = 190.5;
7
8         String [] foods = {"Pizza", "Burger", "Pasta", "Salad"};
9
10        System.out.println("Hello my name is " +name+ " and I am " +age+ " years old.");
11        System.out.println(height + " cm.");
12        System.out.println(x:"My favorite foods are:");
13        System.out.println(foods[0]);
14        System.out.println(foods[1]);
15        System.out.println(foods[2]);
16        System.out.println(foods[3]);
17
18    }
19 }
20
```



# Empty first, fill later - Declared Array

```
J Me
1  import java.util.Arrays;
2  public class MeetSudo {
    Run | Debug
3      public static void main(String []args) {
4          String name = "Sudo";
5          int age = 500;
6
7          double height = 190.5;
8
9          String [] foods = new String[4];
10         foods[0] = "Pizza";
11         foods[1] = "Sushi";
12         foods[2] = "Ice Cream";
13         foods[3] = "Chocolate";
14
15         System.out.println("Hello my name is " + name + " and I am " + age + " years old.");
16         System.out.println(height + " cm.");
17         System.out.println("My favorite foods are:" + Arrays.toString(foods));
18     }
19 }
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Hello my name is Sudo and I am 500 years old.
190.5 cm.
My favorite foods are:[Pizza, Sushi, Ice Cream, Chocolate]
```

# ArrayList

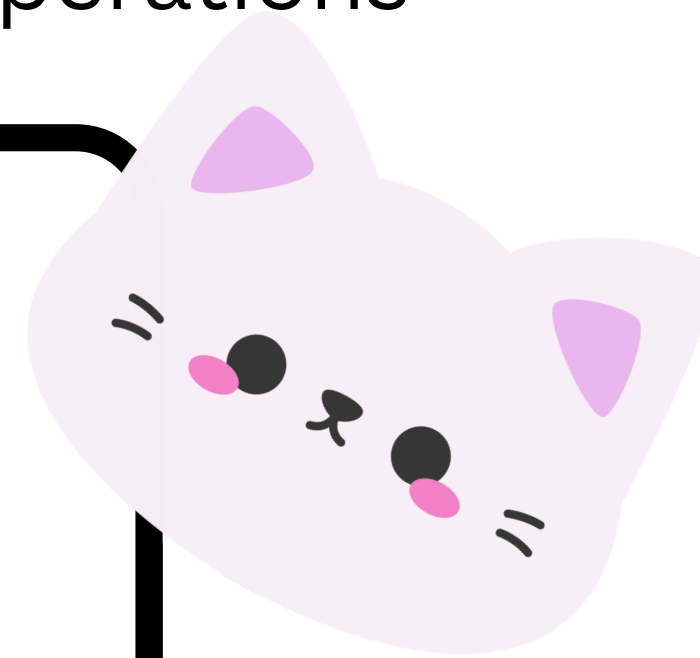
J ArrayListSamp

```
1  import java.util.ArrayList; //library
2  public class ArrayListSample{
    Run | Debug
3      public static void main(String[] args){
4
5          ArrayList<String> menu = new ArrayList<>();
6          menu.add(e:"Ramen"); //adding items
7          menu.add(e:"Sushi");
8          menu.add(e:"Tempura");
9          menu.remove(o:"Tempura"); //deleting items
10         menu.add(e:"Udon");
11         menu.set(menu.indexOf(o:"Udon"), element:"Tonkatsu"); //updating items
12
13         System.out.println(menu); //printing elements
14
15     }
16 }
17
```

# Task

1. Introduce yourself name , age(variable)
2. Favorite colors [3] - Fixed Array
3. Favorite food- (Declared Array) [5] print all values except index 2
4. Dynamic Array [10] (favourite song) crude operations

**Note:** Code this into a single file only class name should be YourfirstnameArray.java  
Comments are optional but recommended



# Task

submit to github repo

Create your own branch  
name it your first name

MottenNutki-1401/  
**childrenofjava**



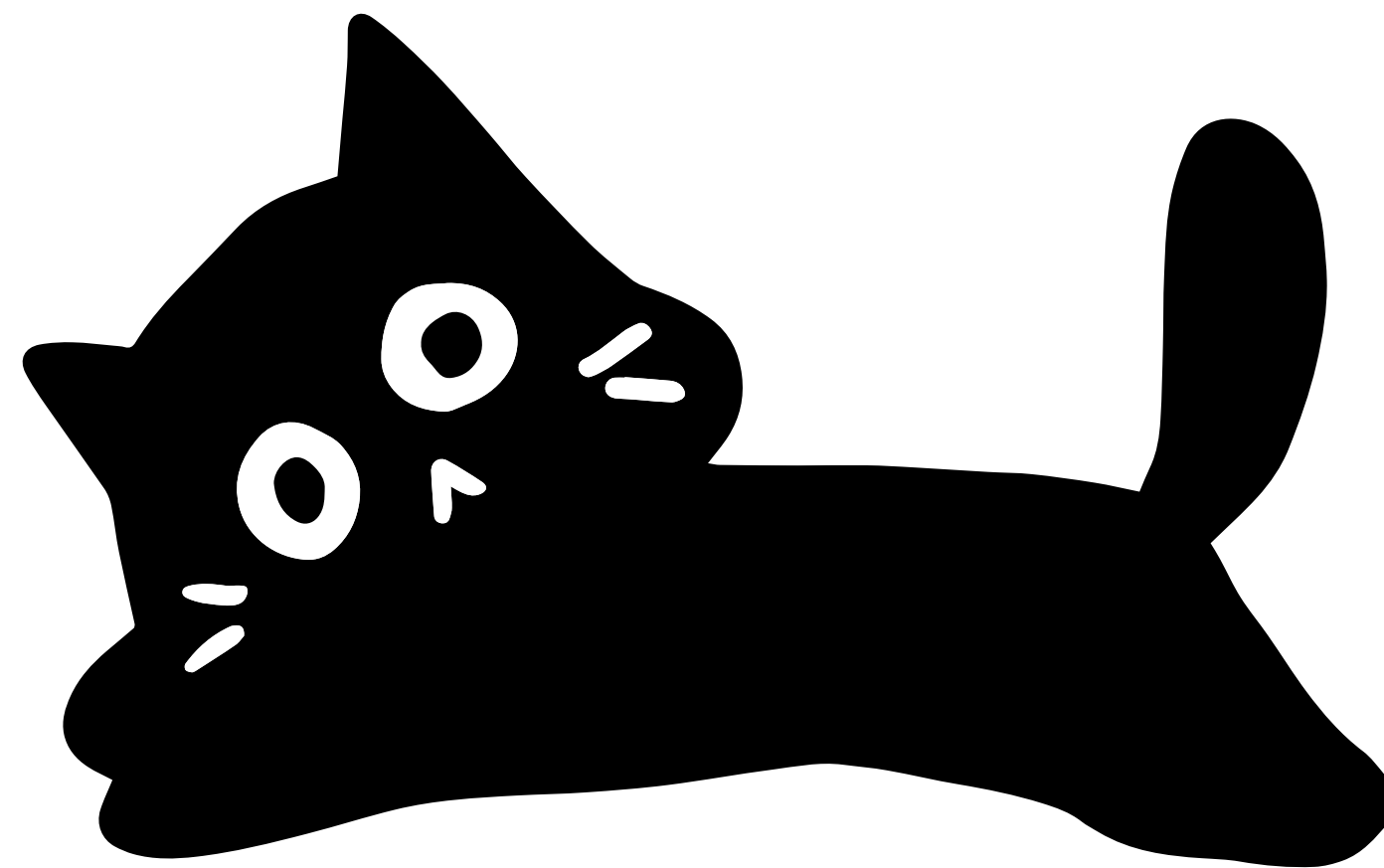
hello children this will be our repository for java DSA

1 Contributor 0 Issues 0 Stars 0 Forks

**MottenNutki-1401/childrenofjava: hello children this will be our repository for java DSA**

hello children this will be our repository for java DSA - MottenNutki-1401/childrenofjava

GitHub



**Thank You!**